# Fractions and LCM: answer key

**A** 1) $4 = 2 \times 2 = 2^2$; $6 = 2 \times 3$: LCM **12** 2) $6 = 2 \times 3$; $9 = 3 \times 3 = 3^2$: LCM **18** 3) $8 = 2 \times 2 \times 2 = 2^3$; $12 = 2 \times 2 \times 3 = 2^2 \times 3$: LCM **24** 4) $10 = 2 \times 5$; $15 = 3 \times 5$: LCM **30** 5) $4 = 2 \times 2 = 2^2$; $6 = 2 \times 3$; $9 = 3 \times 3 = 3^2$: LCM **36** 6) $6 = 2 \times 3$; $10 = 2 \times 5$; $15 = 3 \times 5$: LCM **30**

**B**

1) *same denominator*: LCM 8: $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$

2) *multiple*: LCM 6: $\frac{5}{6} + \frac{2}{6} = \frac{7}{6}$

3) *neither: LCM needed*: LCM 12: $\frac{9}{12} + \frac{2}{12} = \frac{11}{12}$

4) *multiple*: LCM 10: $\frac{7}{10} - \frac{4}{10} = \frac{3}{10}$

5) *neither: LCM needed*: LCM 18: $\frac{10}{18} + \frac{3}{18} = \frac{13}{18}$

6) *neither: LCM needed*: LCM 24: $\frac{22}{24} - \frac{9}{24} = \frac{13}{24}$

**C**

1) LCM 30: $\frac{20}{30} + \frac{9}{30} = \frac{29}{30}$

2) LCM 24: $\frac{21}{24} - \frac{10}{24} = \frac{11}{24}$

3) LCM 30: $\frac{25}{30} - \frac{9}{30} = \frac{16}{30} = \frac{8}{15}$

4) LCM 12: $\frac{9}{12} + \frac{10}{12} = \frac{19}{12}$

5) LCM 45: $\frac{21}{45} + \frac{20}{45} = \frac{41}{45}$

6) LCM 30: $\frac{27}{30} - \frac{14}{30} = \frac{13}{30}$

7) LCM 24: $\frac{10}{24} + \frac{9}{24} = \frac{19}{24}$

8) LCM 30: $\frac{26}{30} - \frac{9}{30} = \frac{17}{30}$

9) LCM 30: $\frac{25}{30} + \frac{14}{30} = \frac{39}{30} = \frac{13}{10}$

**D**

1) Ana added the numerators and the denominators. Denominators must match first: LCM 35, $\frac{21}{35} + \frac{10}{35} = \frac{31}{35}$.

2) Ben changed the denominators to 12 but forgot to multiply the numerators: $\frac{9}{12} - \frac{2}{12} = \frac{7}{12}$.

**E**

1) LCM 12: $\frac{8}{12} + \frac{3}{12} - \frac{2}{12} = \frac{9}{12} = \frac{3}{4}$

2) LCM 24: $\frac{21}{24} - \frac{4}{24} + \frac{6}{24} = \frac{23}{24}$

3) LCM 20: $\frac{15}{20} + \frac{8}{20} - \frac{2}{20} = \frac{21}{20}$

4) LCM 72: $\frac{60}{72} + \frac{27}{72} - \frac{16}{72} = \frac{71}{72}$

**F**

1) LCM 15: $\frac{6}{15} + \frac{5}{15} = \frac{11}{15}$ used; $\frac{15}{15} - \frac{11}{15} = \frac{4}{15}$ left.

2) LCM 24: $\frac{9}{24} + \frac{4}{24} = \frac{13}{24}$ by walk or bus; $\frac{24}{24} - \frac{13}{24} = \frac{11}{24}$ by car.

3) LCM 12: $\frac{3}{12} + \frac{4}{12} + \frac{2}{12} = \frac{9}{12} = \frac{3}{4}$ read; $\frac{12}{12} - \frac{9}{12} = \frac{3}{12} = \frac{1}{4}$ left.

**G**

1) LCM 36: $\frac{21}{36} + \frac{10}{36} - \frac{9}{36} = \frac{22}{36} = \frac{11}{18}$

2) LCM 24: $\frac{15}{24} + \frac{4}{24} - \frac{8}{24} = \frac{11}{24}$

3) LCM 40: $\frac{22}{40} + \frac{15}{40} - \frac{10}{40} = \frac{27}{40}$

4) LCM 30: $\frac{24}{30} + \frac{3}{30} - \frac{5}{30} = \frac{22}{30} = \frac{11}{15}$

★ $\frac{5}{12} - \frac{1}{4} = \frac{5}{12} - \frac{3}{12} = \frac{2}{12} = \frac{1}{6}$. (Check: $\frac{1}{4} + \frac{1}{6} = \frac{3}{12} + \frac{2}{12} = \frac{5}{12}$.)